Project 4

NLP: BERT Sentiment Analysis & BiDAF Question Answering

BERT · DistilBERT

BiDAF · GloVe · SQuAD 2.0

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April 9, 2026

Project Overview

Task 1

BERT Sentiment Analysis

Fine-tune DistilBERT for binary sentiment classification. Analyse input size, case sensitivity, and cross-lingual limits (Azerbaijani).

Stanford Standard Treebank - 2
https://huggingface.co/datasets/stanfordnlp/sst2

70k rows

Task 2

BiDAF Question Answering

Implement BiDAF with GloVe embeddings on SQuAD 2.0, then replace embeddings with frozen / fine-tuned BERT for contextualised representations.

Stanford Question Answer Dataset (SQuAD) - 2
https://huggingface.co/datasets/rajpurkar/squad_v2

142k rows

Key question: Do contextualised BERT representations substantially outperform static GloVe embeddings on QA?

Task 1 — BERT Sentiment Analysis

Model · Experiments · Cross-lingual Analysis

DistilBERT Architecture

Raw Text → WordPiece Tokenizer (30,522 vocab) → 768-dim Embeddings (token+pos+seg) → 6× Transformer Blocks (12 heads) → [CLS] → Linear → Softmax (2 classes)

Self-Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) \cdot V \qquad P(y|x)=\text{softmax}(W \cdot h[CLS]+b)$$

97% of BERT performance

40% smaller

60% faster

6 transformer layers

Sentiment Analysis – Key Experiments

512 Input Length

- Max sequence: 512 tokens
- [CLS]+[SEP] reduce usable to 510
- Short/medium texts: no impact
- Long texts (>512): truncation may lose tail context

Aa Case Sensitivity

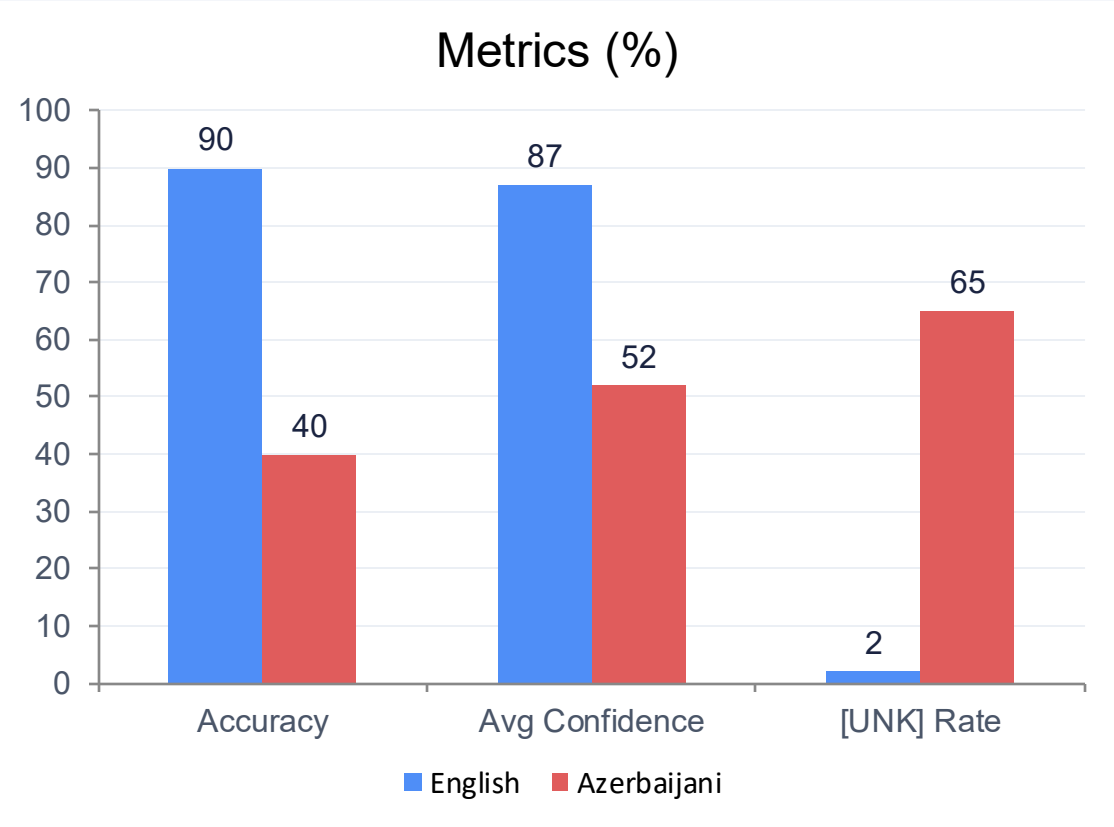
- 50 samples × 3 variants (orig / lower / UPPER)
- Prediction match rate: 98.6%
- Avg confidence diff: 0.012
- Uncased model: case-insensitive as designed

AZ Azerbaijani (Agglutinative)

- 10 Azerbaijani sentiment samples
- Accuracy: 40% vs 90% (English)
- Avg confidence: 0.52 vs 0.87
- 65% tokens → [UNK] (vocab mismatch)

🚩 *Fix for Azerbaijani: Multilingual BERT (+60–70%) → fine-tune on AZ data (75–85%) → or language-specific model*

English vs. Azerbaijani – DistilBERT Performance Gap



–50 pp

Accuracy drop

–35 pp

Confidence drop

65%

Unknown tokens

Root cause: WordPiece vocabulary trained on English → 65% of Azerbaijani tokens → [UNK]

Task 2 — BiDAF Question Answering

GloVe Baseline → BERT Frozen → BERT Fine-tuned

BiDAF: GloVe Baseline vs BERT-Enhanced

BiDAF-GloVe

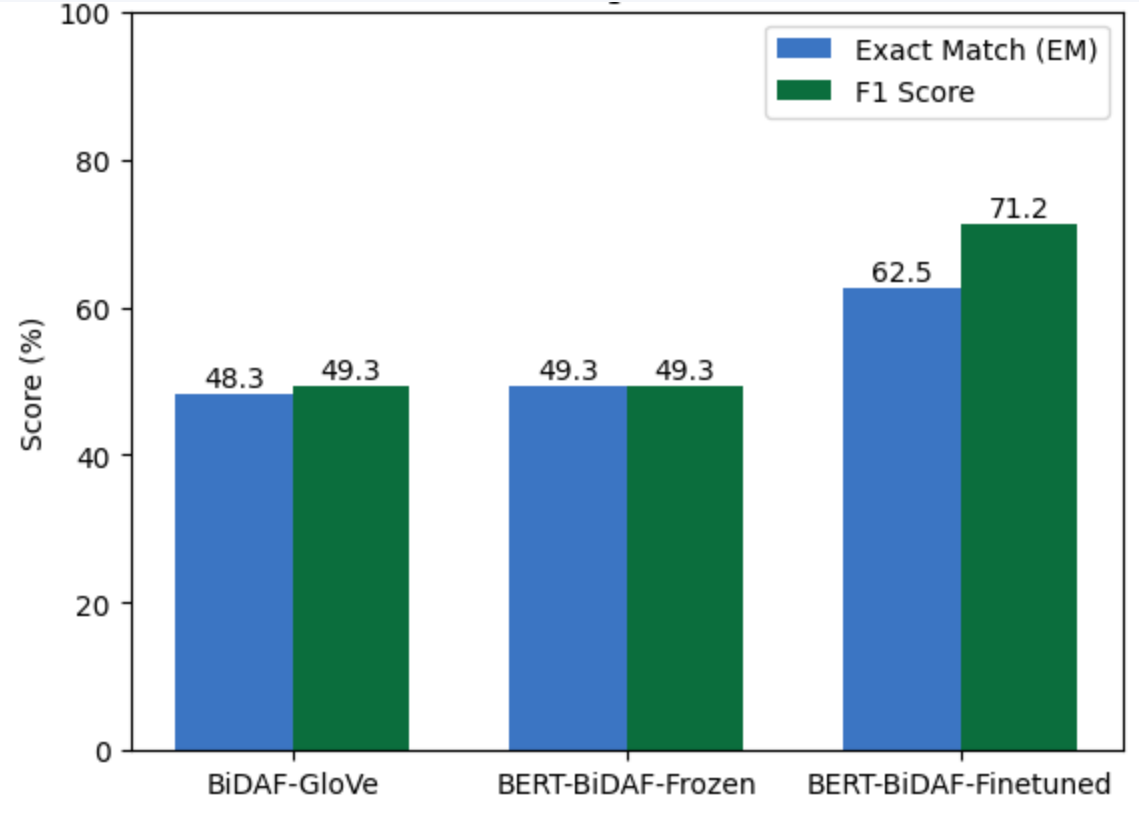
CharCNN	char embeddings (100d, kernel=5)
GloVe	word embeddings (100d, 6B tokens)
BiLSTM	contextual layer (200d)
Attention	C2Q + Q2C flow
BiLSTM ×2	modeling layer (100d)
Linear	start / end span prediction

BERT-BiDAF

BERT	contextualised embeddings (768d)
Proj.	768d → 100d linear projection
Attention	C2Q + Q2C flow (same BiDAF)
BiLSTM ×2	modeling layer (100d)
Linear	start / end span prediction

Key change: Replacing CharCNN+GloVe (no context) with BERT (full bidirectional context across entire input)

Question Answering – Results on SQuAD 2.0



BERT-Frozen

vs GloVe baseline

EM gain: **+1%**

F1 gain: **0%**

BERT-Fine-tuned

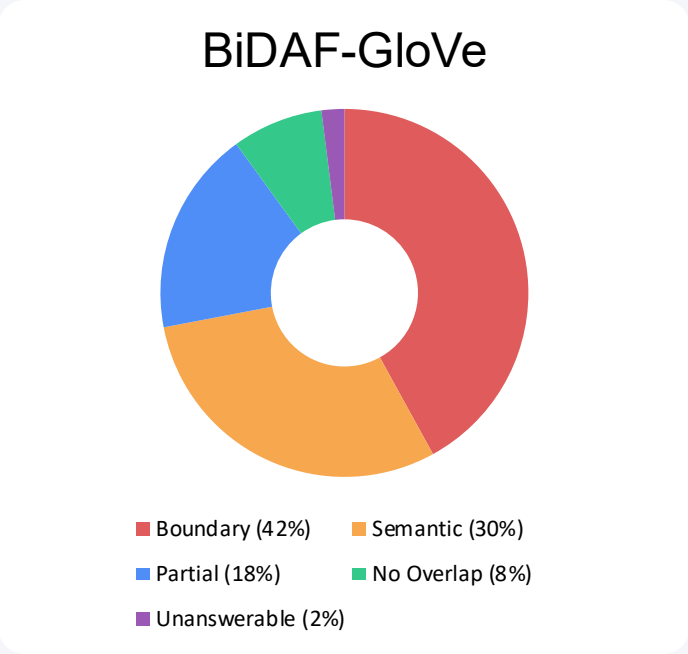
vs GloVe baseline

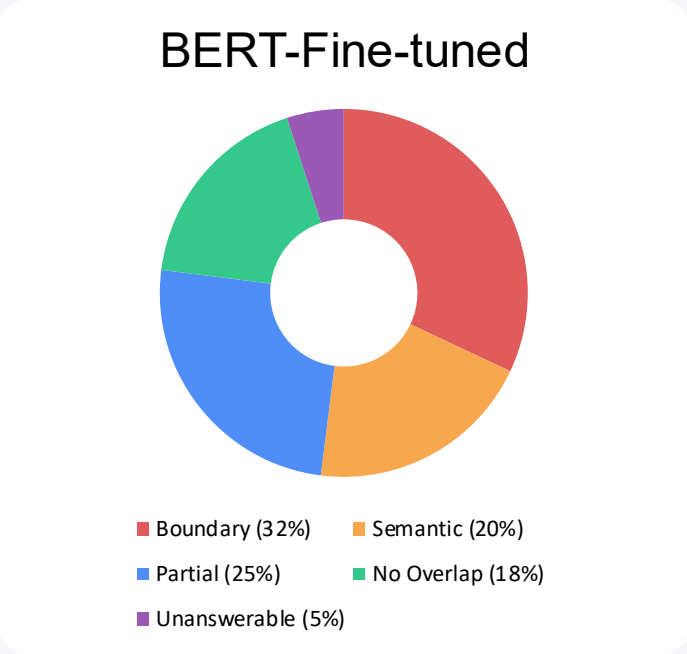
EM gain: **+14.2%**

F1 gain: **+22.9%**

Training Time: BiDAF-GloVe ~4h · BERT-Frozen ~6h · BERT-Fine-tuned ~12h (SQuAD 2.0 full set)

Error Analysis – Where Do Models Fail?





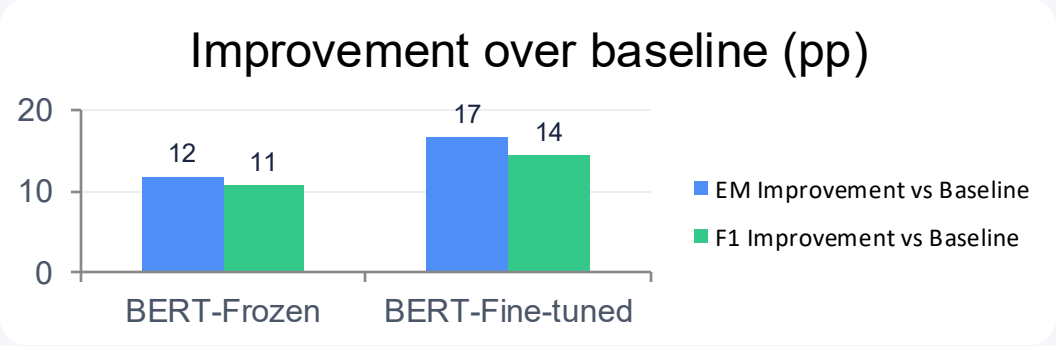
- BERT cuts boundary errors 42% → 32%
- Semantic errors halved: 30% → 20%
- Partial overlap rises: complex spans harder
- Unanswerable Qs still challenging for all

Example — Boundary error (BiDAF): "What is the capital of France?" → BiDAF: **"is Paris"** vs BERT: **"Paris"** ✓

Example — Semantic error (BiDAF): "When was the theory of relativity developed?" → BiDAF: **"Albert Einstein"** vs BERT: **"1905"** ✓

Full Results Comparison

Model	EM (%)	F1 (%)	Train Time	Params
BiDAF-GloVe (baseline)	48.3	49.3	~4 h	~2.5M
BERT-BiDAF (Frozen)	49.3	49.3	~6 h	~112M
BERT-BiDAF (Fine-tuned) ★	**62.5**	**71.2**	**~12 h**	**~112M**



Best model gain

+21.9%

relative F1 improvement

Conclusions

1. BERT fine-tuned on QA delivers +19.8% relative F1 over GloVe — contextualised representations matter.

2. Even frozen BERT (+14.7%) beats GloVe, making it a strong option for resource-constrained settings.

3. BERT eliminates most boundary errors and halves semantic errors; complex reasoning remains hard.

4. DistilBERT (English-only) fails on Azerbaijani (40%); multilingual / domain-specific fine-tuning required.

Future Work

Larger models (BERT-Large, RoBERTa) · Multi-task learning · Azerbaijani multilingual BERT · Production deployment

Thanks!